In the perfectly conducting wire, $\mathbf{D} = \mathbf{E} = 0$; hence, Eq. (6.44) gives $I_{1d} = 0$. As for I_{1c}, we know from circuit theory that it is related to the voltage across the capacitor V_C by

$$\begin{aligned} I_{1c} &= C\frac{dV_C}{dt} \\ &= C\frac{d}{dt}(V_0\cos\omega t) \\ &= -CV_0\omega\sin\omega t, \end{aligned} \tag{6.47}$$

where we used the fact that $V_C = V_s(t)$. With $I_{1d} = 0$, the total current in the wire is simply $I_1 = I_{1c} = -CV_0\omega\sin\omega t$.

In the perfect dielectric with permittivity ε between the capacitor plates, $\sigma = 0$. Hence, $I_{2c} = 0$ because no conduction current exists there. To determine I_{2d}, we need to apply Eq. (6.44). From Example 4-14, the electric field $\mathbf{E}$ in the dielectric spacing is related to the voltage V_c across its plates by

$$\mathbf{E} = \hat{\mathbf{y}}\frac{V_c}{d} = \hat{\mathbf{y}}\frac{V_0}{d}\cos\omega t, \tag{6.48}$$

where d is the spacing between the plates and $\hat{\mathbf{y}}$ is the direction from the higher-potential plate toward the lower-potential plate at $t = 0$. The displacement current I_{2d} is obtained by applying Eq. (6.44) with $d\mathbf{s} = \hat{\mathbf{y}}\,ds$:

$$\begin{aligned} I_{2d} &= \int_S \frac{\partial \mathbf{D}}{\partial t}\cdot d\mathbf{s} \\ &= \int_A \left[\frac{\partial}{\partial t}\left(\hat{\mathbf{y}}\frac{\varepsilon V_0}{d}\cos\omega t\right)\right]\cdot(\hat{\mathbf{y}}\,ds) \\ &= -\frac{\varepsilon A}{d}V_0\omega\sin\omega t \\ &= -CV_0\omega\sin\omega t, \end{aligned} \tag{6.49}$$

where we used the relation $C = \varepsilon A/d$ for the capacitance of the parallel-plate capacitor with plate area A. The expression for I_{2d} in the dielectric region between the conducting plates is identical with that given by Eq. (6.47) for the conduction current I_{1c} in the wire. The fact that these two currents are equal ensures the continuity of total current flow through the circuit.

▶ Even though the displacement current does not transport free charges, it nonetheless **behaves** like a real current. ◀

In the capacitor example, we treated the wire as a perfect conductor, and we assumed that the space between the capacitor plates was filled with a perfect dielectric. If the wire has a finite conductivity σ_w, then $\mathbf{D}$ in the wire would not be zero; therefore, the current I_1 would consist of a conduction current I_{1c} as well as a displacement current I_{1d}; that is, $I_1 = I_{1c} + I_{1d}$. By the same token, if the dielectric spacing material has a nonzero conductivity σ_d, then free charges would flow between the two plates, and I_{2c} would not be zero. In that case, the total current flowing through the capacitor would be $I_2 = I_{2c} + I_{2d}$. No matter the circumstances, the total capacitor current remains equal to the total current in the wire. That is, $I_1 = I_2$.

Example 6-7: Displacement Current Density

The conduction current flowing through a wire with conductivity $\sigma = 2\times10^7$ S/m and relative permittivity $\varepsilon_r = 1$ is given by $I_c = 2\sin\omega t$ (mA). If $\omega = 10^9$ rad/s, find the displacement current.

Solution: The conduction current $I_c = JA = \sigma EA$, where A is the cross section of the wire. Hence,

$$\begin{aligned} E &= \frac{I_c}{\sigma A} = \frac{2\times10^{-3}\sin\omega t}{2\times10^7 A} \\ &= \frac{1\times10^{-10}}{A}\sin\omega t \qquad \text{(V/m)}. \end{aligned}$$

Application of Eq. (6.44), with $D = \varepsilon E$, leads to

$$\begin{aligned} I_d &= J_d A \\ &= \varepsilon A\frac{\partial E}{\partial t} \\ &= \varepsilon A\frac{\partial}{\partial t}\left(\frac{1\times10^{-10}}{A}\sin\omega t\right) \\ &= \varepsilon\omega\times10^{-10}\cos\omega t \\ &= 0.885\times10^{-12}\cos\omega t \quad \text{(A)}, \end{aligned}$$

where we used $\omega = 10^9$ rad/s and $\varepsilon = \varepsilon_0 = 8.85\times10^{-12}$ F/m. Note that I_c and I_d are in phase quadrature (90° phase shift between them). Also, I_d is about nine orders of magnitude smaller than I_c, which is why the displacement current usually is ignored in good conductors.

Exercise 6-5: A poor conductor is characterized by a conductivity $\sigma = 100$ (S/m) and permittivity $\varepsilon = 4\varepsilon_0$. At what angular frequency ω is the amplitude of the conduction current density $\mathbf{J}$ equal to the amplitude of the displacement current density $\mathbf{J}_d$?

Answer: $\omega = 2.82\times10^{12}$ (rad/s). (See EM.)

在理想导线中，D = E = 0；因此，等式(6.44) 给出 I 1d = 0。至于 I1c，我们从电路理论得知，它与电容器 VC 两端的电压有关：

$$\begin{aligned} I_{1c} &= C\frac{dV_C}{dt} \\ &= C\frac{d}{dt}(V_0\cos\omega t) \\ &= -CV_0\omega\sin\omega t, \end{aligned} \tag{6.47}$$

我们使用了 VC = Vs(t) 这一事实。当 I1d = 为 0 时，导线中的总电流就是 I1 = I1c = −CV0ω sin ωt。

在电容器板之间介电常数为 ε 的理想电介质中，σ = 为 0。因此，I2c = 为 0，因为那里不存在传导电流。为了确定 I2d，我们需要应用等式： (6.44) 。从例 4-14 可以看出，电介质间距中的电场 E 与其极板上的电压 Vc 的关系为：

$$\mathbf{E} = \hat{\mathbf{y}}\frac{V_c}{d} = \hat{\mathbf{y}}\frac{V_0}{d}\cos\omega t, \tag{6.48}$$

其中 d 是极板之间的间距，ŷ 是在 t = 0 时从较高电势极板朝向较低电势极板的方向。位移电流 I2d 通过应用等式获得。 (6.44) 与 ds = ŷ ds:

$$\begin{aligned} I_{2d} &= \int_S \frac{\partial \mathbf{D}}{\partial t}\cdot d\mathbf{s} \\ &= \int_A \left[\frac{\partial}{\partial t}\left(\hat{\mathbf{y}}\frac{\varepsilon V_0}{d}\cos\omega t\right)\right]\cdot(\hat{\mathbf{y}}\,ds) \\ &= -\frac{\varepsilon A}{d}V_0\omega\sin\omega t \\ &= -CV_0\omega\sin\omega t, \end{aligned} \tag{6.49}$$

其中，我们使用关系式 C = ε A/d 来表示板面积为 A 的平行板电容器的电容。导电板之间介电区域中 I2d 的表达式与等式 1 给出的相同。 (6.47) 为导线中的传导电流 I1c 。这两个电流相等的事实确保了流过电路的总电流的连续性。

▶ 尽管位移电流不传输自由电荷，但它的行为仍然像真实电流一样。 ◀

在电容器的例子中，我们将电线视为理想的导体，并假设电容器板之间的空间充满了理想的电介质。如果电线具有有限电导率 σw，则导线中的 D 不会为零；因此，电流 I1 将由传导电流 I1c 和位移电流 I1d 组成；也就是说，I 1 = I1c + I1d。出于同样的原因，如果介电间隔材料具有非零电导率 σd，则自由电荷将在两个板之间流动，并且 I 2c 将不为零。在这种情况下，流过电容器的总电流将为 I 2 = I2c + I2d。无论情况如何，总电容器电流始终等于电线中的总电流。也就是说，我1 =我2。

例 6-7： 位移电流密度

流经电导率 σ = 2 × 10^7 S/m 和相对介电常数 εr = 1 的导线的传导电流由 Ic = 2 sin ωt (mA) 给出。如果 ω = 10^9 rad/s，求位移电流。

解：传导电流Ic =JA= σ EA，其中A是导线的横截面。因此，

$$\begin{aligned} E &= \frac{I_c}{\sigma A} = \frac{2\times10^{-3}\sin\omega t}{2\times10^7 A} \\ &= \frac{1\times10^{-10}}{A}\sin\omega t \qquad \text{(V/m)}。 \end{aligned}$$

方程的应用（6.44），与 D = ε E，导致

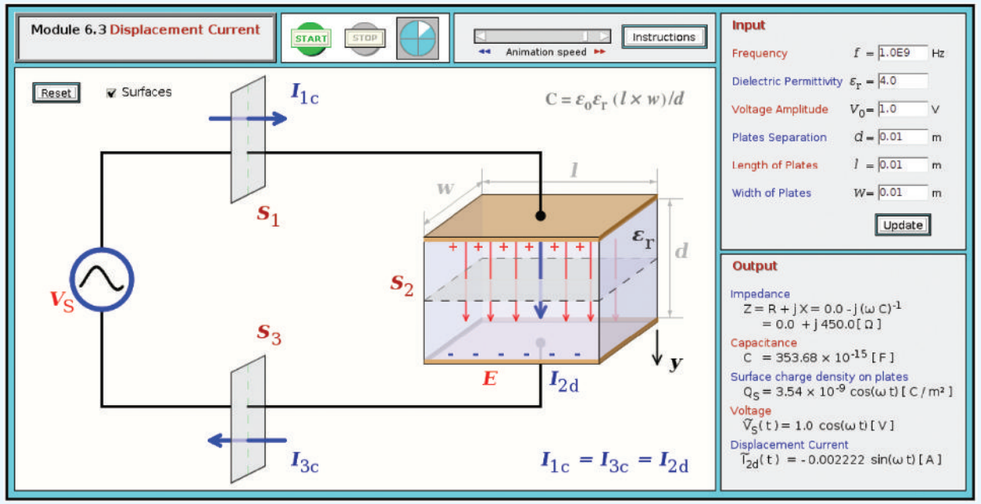
$$\begin{aligned} I_d &= J_d A \\ &= \varepsilon A\frac{\partial E}{\partial t} \\ &= \varepsilon A\frac{\partial}{\partial t}\left(\frac{1\times10^{-10}}{A}\sin\omega t\right) \\ &= \varepsilon\omega\times10^{-10}\cos\omega t \\ &= 0.885\times10^{-12}\cos\omega t \quad \text{(A)}, \end{aligned}$$

其中我们使用 ω = 10^9 rad/s 和 ε = ε0 = 8.85 × 10^−12 F/m 。请注意，Ic 和 Id 相位正交（它们之间有 90° 相移）。此外，Id 比 Ic 小约 9 个数量级，这就是为什么在良导体中位移电流通常被忽略的原因。

练习 6-5：不良导体的特征是电导率 σ = 100 (S/m) 和介电常数 ε = 4ε0。在什么角频率ω处，传导电流密度J的幅值等于位移电流密度Jd的幅值?

答案：ω = 2.82 × 10^12 (rad/s)。 （参见EM。）

Module 6.3 Displacement Current Observe the displacement current through a parallel-plate capacitor.



6-8 Boundary Conditions for Electromagnetics

In Chapters 4 and 5, we applied the integral form of Maxwell's equations under static conditions to obtain boundary conditions applicable to the tangential and normal components of **E**, **D**, **B**, and **H** on interfaces between contiguous media (Section 4-8 for **E** and **D** and in Section 5-6 for **B** and **H**). In the dynamic case, Maxwell's equations (**Table 6-1**) include two new terms not accounted for in electrostatics and magnetostatics, namely, $\partial \mathbf{B}/\partial t$ in Faraday's law and $\partial \mathbf{D}/\partial t$ in Ampère's law.

► Nevertheless, the boundary conditions derived previously for electrostatic and magnetostatic fields remain valid for time-varying fields as well. ◄

This is because, if we were to apply the procedures outlined in the previously referenced sections for time-varying fields, we would find that the combination of the aforementioned terms vanish as the areas of the rectangular loops in **Figs. 4-20** and **5-24** are made to approach zero.

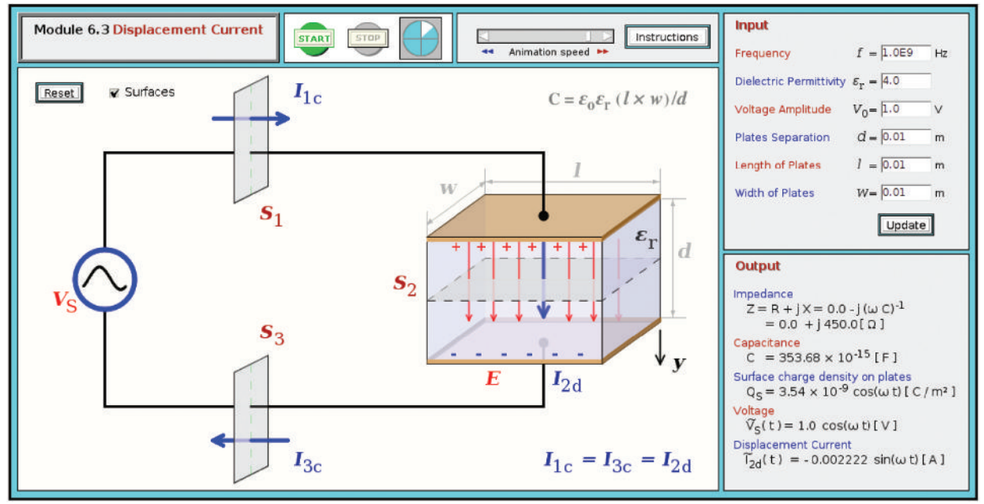
The combined set of electromagnetic boundary conditions is summarized in **Table 6-2**.

Concept Question 6-9: When conduction current flows through a material, a certain number of charges enter the material on one end and an equal number leave on the other end. What's the situation like for the displacement current through a perfect dielectric?

Concept Question 6-10: Verify that the integral form of Ampère's law given by Eq. (6.43) leads to the boundary condition that the tangential component of **H** is continuous across the boundary between two dielectric media.

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模块 6.3 位移电流 观察通过平行板电容器的位移电流。



6-8 电磁学的边界条件

在第4章和第5章中，我们在静态条件下应用了麦克斯韦方程组的积分形式，以获得适用于连续介质之间的界面上的E、D、B和H的切向和法向分量的边界条件（E和D在4-8节中，B和H在5-6节中）。在动态情况下，麦克斯韦方程组（表 6-1）包括静电学和静磁学中未考虑的两个新项，即法拉第定律中的 ∂B/∂t 和安培定律　　中的 ∂D/∂t。

► 然而，先前针对静电场和静磁场导出的边界条件对于时变场仍然有效。 ◄

这是因为，如果我们将前面引用的部分中概述的过程应用于时变场，我们会发现上述术语的组合随着图 2 和 3 中矩形环的面积而消失。使4-20和5-24接近于零。

表 6-2 总结了电磁边界条件的组合集。

概念问题 6-9：当传导电流流过材料时，一定数量的电荷从一端进入材料，并在另一端离开相同数量的电荷。通过理想电介质的位移电流是什么样的情况?

概念问题 6-10：验证方程 1 给出的安培定律　　的积分形式。 (6.43) 得出这样的边界条件：H 的切向分量在两个介电介质之间的边界上是连续的。

Table 6-2 Boundary conditions for the electric and magnetic fields.

Field Components	General Form	Medium 1 Dielectric / Medium 2 Dielectric	Medium 1 Dielectric / Medium 2 Conductor
Tangential E	$\hat{\mathbf{n}}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$	$E_{1t} = E_{2t}$	$E_{1t} = E_{2t} = 0$
Normal D	$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$ / $D_{2n} = 0$
Tangential H	$\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$	$H_{1t} = H_{2t}$	$H_{1t} = J_s$ / $H_{2t} = 0$
Normal B	$\hat{\mathbf{n}}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$B_{1n} = B_{2n}$	$B_{1n} = B_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{\mathbf{n}}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.

6-9 Charge–Current Continuity Relation

Under static conditions, the charge density ρ_v and the current density $\mathbf{J}$ at a given point in a material are totally independent of one another. This is no longer true in the time-varying case. To show the connection between ρ_v and $\mathbf{J}$, we start by considering an arbitrary volume υ bounded by a closed surface S (Fig. 6-14). The net positive charge contained in υ is Q. Since, according to the law of conservation of electric charge (Section 1-3.2), charge can neither be created nor destroyed, the only way Q can increase is as a result of a net inward flow of positive charge into the volume υ. By the same token, for Q to decrease, there has to be a net outward flow of charge from υ. The inward and outward flow of charge constitute currents flowing across the surface S into and out of υ, respectively. We define I as the *net current flowing across S out of υ*. Accordingly, I is equal to the *negative* rate of change of Q:

$$I = -\frac{dQ}{dt} = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon, \tag{6.50}$$

where ρ_v is the volume charge density in υ. According to Eq. (4.12), the current I is also defined as the outward flux of the current density $\mathbf{J}$ through the surface S. Hence,

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon. \tag{6.51}$$

By applying the divergence theorem given by Eq. (3.98), we can convert the surface integral of $\mathbf{J}$ into a volume integral of its divergence $\nabla \cdot \mathbf{J}$, which then gives

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_{\upsilon} \nabla \cdot \mathbf{J} \, d\upsilon = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon. \tag{6.52}$$

For a stationary volume υ, the time derivative operates on ρ_v only. Hence, we can move it inside the integral and express it as a partial derivative of ρ_v:

$$\int_{\upsilon} \nabla \cdot \mathbf{J} \, d\upsilon = -\int_{\upsilon} \frac{\partial \rho_v}{\partial t} \, d\upsilon. \tag{6.53}$$

In order for the volume integrals on both sides of Eq. (6.53) to be equal for any volume υ, their integrands have to be equal at every point within υ. Hence,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}, \tag{6.54}$$

which is known as the ***charge–current continuity relation***, or simply the ***charge continuity equation***.

If the volume charge density within an elemental volume $\Delta\upsilon$ (such as a small cylinder) is not a function of time

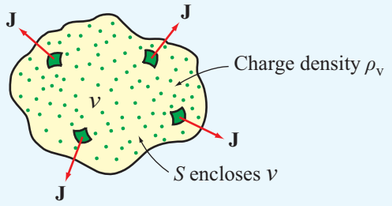


Figure 6-14 The total current flowing out of a volume υ is equal to the flux of the current density $\mathbf{J}$ through the surface S, which in turn is equal to the rate of decrease of the charge enclosed in υ.

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表 6-2 电场和磁场的边界条件。

Field Components	General Form	Medium 1 Dielectric / Medium 2 Dielectric	Medium 1 Dielectric / Medium 2 Conductor
Tangential E	$\hat{\mathbf{n}}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$	$E_{1t} = E_{2t}$	$E_{1t} = E_{2t} = 0$
Normal D	$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$ / $D_{2n} = 0$
Tangential H	$\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$	$H_{1t} = H_{2t}$	$H_{1t} = J_s$ / $H_{2t} = 0$
Normal B	$\hat{\mathbf{n}}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$B_{1n} = B_{2n}$	$B_{1n} = B_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{\mathbf{n}}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.

6-9 充电-电流连续性关系

在静态条件下，材料中给定点的电荷密度ρ_v和电流密度J完全彼此独立。在随时间变化的情况下，情况不再如此。为了显示 ρ_v 和 J 之间的联系，我们首先考虑由封闭曲面 S 界定的任意体积 υ （图 6-14）。υ 中包含的净正电荷为 Q。根据电荷守恒定律（第 1-3.2 节），电荷既不能产生也不能消失，因此 Q 增加的唯一方法是正电荷净向内流入体积 υ。出于同样的原因，为了使 Q 减小，必须有来自 υ 的净向外电荷流。向内和向外的电荷流分别构成穿过表面 S 流入和流出 υ 的电流。我们将 I 定义为 υ 中流经 S 的净电流。因此，I 等于负利率

Q 的变化：

$$I = -\frac{dQ}{dt} = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon, \tag{6.50}$$

其中 ρ_v 是 υ 中的体积电荷密度。根据方程。(4.12) 中，电流 I 也定义为通过表面 S 的电流密度 J 的向外通量。因此，

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon. \tag{6.51}$$

通过应用方程给出的散度定理。(3.98)，我们可以将 J 的表面积分转换为其散度 $\nabla \cdot$ J 的体积积分，然后得到

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_{\upsilon} \nabla \cdot \mathbf{J} \, d\upsilon = -\frac{d}{dt}\int_{\upsilon} \rho_v \, d\upsilon. \tag{6.52}$$

对于固定体积 υ，时间导数仅对 ρ_v 起作用。因此，我们可以将其移入积分并将其表示为 ρ_v 的偏导数：

$$\int_{\upsilon} \nabla \cdot \mathbf{J} \, d\upsilon = -\int_{\upsilon} \frac{\partial \rho_v}{\partial t} \, d\upsilon. \tag{6.53}$$

为了求方程两边的体积积分：(6.53) 对于任何体积 υ 都相等，它们的被积数必须在 υ 内的每个点都相等。因此，

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}, \tag{6.54}$$

这称为电荷-电流连续性关系，或简称为电荷连续性方程。

如果元素体积 $\Delta\upsilon$（（例如小圆柱体））内的体积电荷密度不是时间的函数

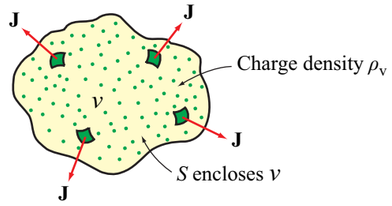


图 6-14 从体积 υ 流出的总电流等于通过表面 S 的电流密度 J 的通量，而该通量又等于 υ 中所包含的电荷的减少率。

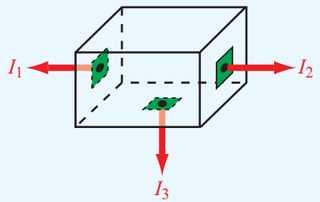


Figure 6-15 Kirchhoff's current law states that the algebraic sum of all the currents flowing out of a junction is zero.

(i.e., $\partial\rho_v/\partial t = 0$), it means that the net current flowing out of $\Delta\mathcal{U}$ is zero or, equivalently, that the current flowing into $\Delta\mathcal{U}$ is equal to the current flowing out of it. In this case, Eq. (6.54) implies

$$\nabla\cdot\mathbf{J} = 0, \tag{6.55}$$

and its integral-form equivalent [from Eq. (6.51)] is

$$\oint_S \mathbf{J}\cdot d\mathbf{s} = 0. \quad \textbf{(Kirchhoff's current law)} \tag{6.56}$$

Let us examine the meaning of Eq. (6.56) by considering a junction (or node) connecting two or more branches in an electric circuit. No matter how small, the junction has a volume $\mathcal{U}$ enclosed by a surface S. The junction shown in **Fig. 6-15** has been drawn as a cube, and its dimensions have been artificially enlarged to facilitate the present discussion. The junction has six faces (surfaces), which collectively constitute the surface S associated with the closed-surface integration given by Eq. (6.56). For each face, the integration represents the current flowing out through that face. Thus, Eq. (6.56) can be cast as

$$\sum_i I_i = 0, \quad \textbf{(Kirchhoff's current law)} \tag{6.57}$$

where I_i is the current flowing outward through the ith face. For the junction of **Fig. 6-15**, Eq. (6.57) translates into $(I_1 + I_2 + I_3) = 0$. In its general form, Eq. (6.57) is an expression of ***Kirchhoff's current law***, which states that in an electric circuit *the sum of all the currents flowing out of a junction is zero.*

6-10 Free-Charge Dissipation in a Conductor

We stated earlier that current flow in a conductor is realized by the movement of loosely attached electrons under the influence of an externally applied electric field. These electrons, however, *are not excess charges*; their charge is balanced by an equal amount of positive charge in the atoms' nuclei. In other words, the conductor material is electrically neutral, and the net charge density in the conductor is zero ($\rho_v = 0$). What happens then if an excess free charge q is introduced at some interior point in a conductor? The excess charge gives rise to an electric field, which forces the charges of the host material nearest to the excess charge to rearrange their locations, which in turn cause other charges to move, and so on. The process continues until neutrality is reestablished in the conductor material and a charge equal to q resides on the conductor's surface.

How fast does the excess charge dissipate? To answer this question, let us introduce a volume charge density ρ_{vo} at the interior of a conductor and then find out the rate at which it decays down to zero. From Eq. (6.54), the continuity equation is given by

$$\nabla\cdot\mathbf{J} = -\frac{\partial\rho_v}{\partial t}. \tag{6.58}$$

In a conductor, the point form of Ohm's law given by Eq. (4.73) states that $\mathbf{J} = \sigma\mathbf{E}$. Hence,

$$\sigma\nabla\cdot\mathbf{E} = -\frac{\partial\rho_v}{\partial t}. \tag{6.59}$$

Next, we use Eq. (6.1), $\nabla\cdot\mathbf{E} = \rho_v/\varepsilon$, to obtain the partial differential equation

$$\frac{\partial\rho_v}{\partial t} + \frac{\sigma}{\varepsilon}\rho_v = 0. \tag{6.60}$$

Given that $\rho_v = \rho_{vo}$ at $t = 0$, the solution of Eq. (6.60) is

$$\rho_v(t) = \rho_{vo}e^{-(\sigma/\varepsilon)t} = \rho_{vo}e^{-t/\tau_r} \qquad (\text{C/m}^3), \tag{6.61}$$

where $\tau_r = \varepsilon/\sigma$ is called the ***relaxation time constant***. We see from Eq. (6.61) that the initial excess charge ρ_{vo} decays exponentially at a rate τ_r. At $t = \tau_r$, the initial charge ρ_{vo} will have decayed to $1/e \approx 37\%$ of its initial value, and at $t = 3\tau_r$, it will have decayed to $e^{-3} \approx 5\%$ of its initial value at $t = 0$. For copper, with $\varepsilon \approx \varepsilon_0 = 8.854\times10^{-12}$ F/m and $\sigma = 5.8\times10^7$ S/m, $\tau_r = 1.53\times10^{-19}$ s. Thus, the charge dissipation process in a conductor is extremely fast. In contrast, the decay rate is very slow in a good insulator. For a material like mica with $\varepsilon = 6\varepsilon_0$ and $\sigma = 10^{-15}$ S/m, $\tau_r = 5.31\times10^4$ s, or approximately 14.8 hours.

Concept Question 6-11: Explain how the charge continuity equation leads to Kirchhoff's current law.

Concept Question 6-12: How long is the relaxation time constant for charge dissipation in a perfect conductor? In a perfect dielectric?

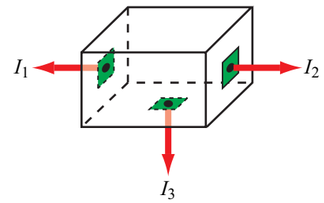


图 6-15 基尔霍夫电流定律指出，流出结点的所有电流的代数和为零。

（IE。，$\partial\rho_v/\partial t=0$），这意味着流出的净电流$\Delta\mathcal{U}$为零，或者等价地，流入的电流$\Delta\mathcal{U}$等于从中流出的电流。在这种情况下，等式。(6.54) 意味着

$$\nabla\cdot\mathbf{J} = 0, \tag{6.55}$$

及其积分形式等价物[来自方程。(6.51)]是

$$\oint_S \mathbf{J}\cdot d\mathbf{s} = 0. \quad \textbf{(Kirchhoff's current law)} \tag{6.56}$$

让我们检查一下等式的含义。（6.56）通过考虑连接电路中两个或多个分支的结点（或节点）。不管结有多小，其体积$\mathcal{U}$都被表面S包围。图6-15中所示的结被画成立方体，并且其尺寸被人为地放大以方便当前的讨论。连接点有六个面（表面），它们共同构成了与方程（1）给出的闭表面积分相关的表面 S。（6.56）。对于每个面，积分代表通过该面流出的电流。因此，等式。(6.56) 可以表示为

$$\sum_i I_i = 0, \quad \textbf{(Kirchhoff's current law)} \tag{6.57}$$

其中 I_i 是通过第 i 个面向外流动的电流。对于图 6-15 的连接点，方程式(6.57) 转化为 $(I_1 + I_2 + I_3) = 0$。在其一般形式中，方程：（6.57）是基尔霍夫电流定律的表达式，它指出在电路中流出结点的所有电流之和为零。

6-10 导体中的自由电荷耗散

我们之前说过，导体中的电流是通过内部松散附着的电子的运动来实现的。

外部施加的电场的通量。然而，这些电子并不是多余的电荷；而是多余的电荷。它们的电荷由原子核中等量的正电荷平衡。换句话说，导体材料是电中性的，导体中的净电荷密度为零（$\rho_v = 0$）。如果在导体的某个内部点引入过量的自由电荷 q，会发生什么情况？多余的电荷会产生电场，迫使最靠近多余电荷的主体材料的电荷重新排列其位置，进而导致其他电荷移动，依此类推。该过程持续进行，直到导体材料重新建立中性并且导体表面上存在等于 q 的电荷。

多余电荷消散的速度有多快？为了回答这个问题，让我们在导体内部引入体积电荷密度 ρ_{vo}，然后找出它衰减到零的速率。从方程。(6.54)，连续性方程由下式给出

$$\nabla\cdot\mathbf{J} = -\frac{\partial\rho_v}{\partial t}. \tag{6.58}$$

在导体中，欧姆定律的点形式由方程式给出。(4.73) 指出 $\mathrm{J} = \sigma\mathrm{E}$。因此，

$$\sigma\nabla\cdot\mathbf{E} = -\frac{\partial\rho_v}{\partial t}. \tag{6.59}$$

接下来，我们使用等式。(6.1)、$\nabla\cdot\mathrm{E} = \rho_v/\varepsilon$，得到偏微分方程

$$\frac{\partial\rho_v}{\partial t} + \frac{\sigma}{\varepsilon}\rho_v = 0. \tag{6.60}$$

鉴于 $t = 0$ 时的 $\rho_v = \rho_{vo}$，方程的解为：(6.60) 是

$$\rho_v(t) = \rho_{vo}e^{-(\sigma/\varepsilon)t} = \rho_{vo}e^{-t/\tau_r} \qquad (\text{C/m}^3), \tag{6.61}$$

其中 $\tau_r = \varepsilon/\sigma$ 称为弛豫时间常数。我们从等式中看到。（6.61）初始过量电荷ρ_{vo}以速率τ_r呈指数衰减。在 $t = \tau_r$ 处，初始电荷 ρ_{vo} 将衰减至其初始值的 $1/\mathrm{e} \approx 37\%$，在 $t = 3\tau_r$ 处，它将衰减至其在 $t = 0$ 处初始值的 $\mathrm{e}^{-3} \approx 5\%$。对于铜，$\varepsilon \approx \varepsilon_0 = 8.854\times10^{-12}$ F/m 和 $\sigma = 5.8\times10^7$ S/m、$\tau_r = 1.53\times10^{-19}$ 秒。因此，导体中的电荷耗散过程非常快。相反，良好绝缘体的衰减率非常慢。对于像云母这样的材料，$\varepsilon = 6\varepsilon_0$ 和 $\sigma = 10^{-15}$ S/m、$\tau_r = 5.31\times10^4$ s 或大约 14.8 小时。

概念问题 6-11：解释电荷连续性方程如何推导出基尔霍夫电流定律。

概念问题 6-12：理想导体中电荷耗散的弛豫时间常数是多长？在理想的电介质中？

Technology Brief 12: EMF Sensors

An ***electromotive force*** (emf) sensor is a device that can generate an induced voltage in response to an external stimulus. Three types of emf sensors are profiled in this technical brief: the ***piezoelectric transducer***, the ***Faraday magnetic flux sensor***, and the ***thermocouple***.

Piezoelectric Transducers

► ***Piezoelectricity*** is the property exhibited by certain crystals, such as quartz, that become electrically polarized when the crystal is subjected to mechanical pressure, thereby inducing a voltage across it. ◄

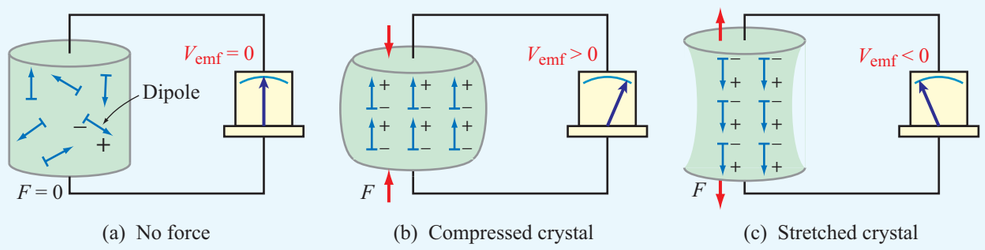
The crystal consists of polar domains represented by equivalent dipoles (**Fig. TF12-1**). Under the absence of an external force, the polar domains are randomly oriented throughout the material, but when ***compressive*** or ***tensile*** (stretching) stress is applied to the crystal, the polar domains align themselves along one of the principal axes of the crystal, leading to a net polarization (electric charge) at the crystal surfaces. Compression and stretching generate voltages of opposite polarity. The piezoelectric effect (***piezein*** means to press or squeeze in Greek) was discovered by the ***Curie brothers***, Pierre and Paul-Jacques, in 1880, and a year later, Lippmann predicted the converse property: If subjected to an electric field, the crystal would change in shape.

► The piezoelectric effect is a ***reversible (bi-directional)*** electromechanical process; application of force induces a voltage across the crystal, and conversely, application of a voltage changes the shape of the crystal. ◄

Piezoelectric crystals are used in ***microphones*** to convert mechanical vibrations (of the crystal surface) caused by acoustic waves into a corresponding electrical signal, and the converse process is used in ***loudspeakers*** to convert electrical signals into sound. In addition to having stiffness values comparable to that of steel, some piezoelectric materials exhibit very high sensitivity to the force applied upon them, with excellent linearity over a wide dynamic range. They can be used to measure surface deformations as small as ***nanometers*** (10^{-9} m), making them particularly attractive as positioning sensors in ***scanning tunneling microscopes***. As ***accelerometers***, they can measure acceleration levels as low as 10^{-4} g to as high as 100 g (where g is the acceleration due to gravity). Piezoelectric crystals and ceramics are used in cigarette lighters and gas grills as spark generators, in clocks and electronic circuitry as precision oscillators, in medical ***ultrasound*** diagnostic equipment as transducers (**Fig. TF12-2**), and in numerous other applications.

Faraday Magnetic Flux Sensor

According to Faraday's law [Eq. (6.6)], the emf voltage induced across the terminals of a conducting loop is directly proportional to the time rate of change of the



(a) No force (b) Compressed crystal (c) Stretched crystal

Figure TF12-1 Response of a piezoelectric crystal to an applied force.

技术简介 12：EMF 传感器

电动势（emf）传感器是一种可以响应外部刺激而产生感应电压的装置。本技术简介介绍了三种类型的电动势传感器：压电传感器、法拉第磁通量传感器和热电偶。

压电换能器

► 压电是某些晶体（例如石英）所表现出的特性，当晶体受到机械压力时，晶体会发生电极化，从而在晶体上感应出电压。 ◄

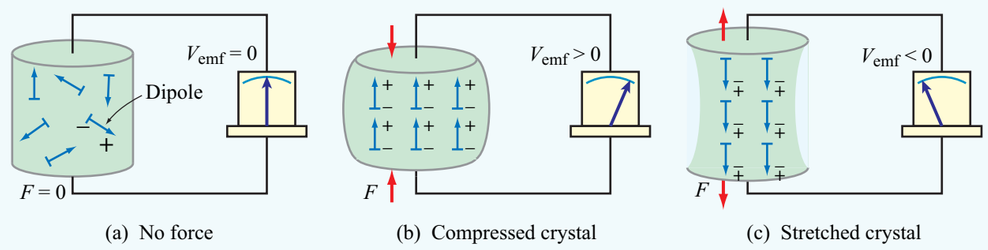
该晶体由由等效偶极子表示的极性域组成（图TF12-1）。在没有外力的情况下，极域在整个材料中随机取向，但当向晶体施加压缩或拉伸应力时，极域沿着晶体的主轴之一排列，导致晶体表面出现净极化（电荷）。压缩和拉伸会产生相反极性的电压。压电效应（piezein 在希腊语中的意思是按压或挤压）是由居里兄弟 Pierre 和 Paul -Jacques 于 1880 年发现的，一年后，Lippmann 预测了相反的特性：如果受到电场的影响，晶体的形状会发生变化。

► 压电效应是一种可逆（双向）机电过程；施加力会在晶体上感应出电压，相反，施加电压会改变晶体的形状。 ◄

压电晶体用于麦克风中，将声波引起的（晶体表面的）机械振动转换为相应的电信号，而相反的过程则用于扬声器中，将电信号转换为声音。除了具有与钢相当的刚度值之外，一些压电材料还对施加在其上的力表现出非常高的灵敏度，并且在宽动态范围内具有出色的线性度。它们可用于测量小至纳米（10^{-9}m）的表面变形，这使得它们作为扫描隧道显微镜中的定位传感器特别有吸引力。作为加速度计，它们可以测量低至 10^{-4} g 至高至 100 g 的加速度水平（其中 g 是重力加速度）。压电晶体和陶瓷在打火机和燃气烤架中用作火花发生器，在时钟和电子电路中用作精密振荡器，在医疗超声诊断设备中用作传感器（图TF12-2），以及许多其他应用。

法拉第磁通传感器

根据法拉第定律[Eq. (6.6)]，导电回路两端感应的电动势电压与电流变化率的时间成正比。



(a) No force (b) Compressed crystal (c) Stretched crystal

图 TF12-1 压电晶体对所施加的力的响应。